Actual finite extensions and Frobenius iterates

For actual finite fields F⊆K, a field extension B/K and a general Weierstrass model W over F, Lean proves the following actual-point equivalence. Here $q = |F|$, $k = [K : F]$ and the scalar-tower embeddings are explicit. [pcc:unverified]

$$\mathrm{Frob}_q^k(P) = P \iff \exists Q \in W(K) : \mathrm{map}_{K \to B}(Q) = P.$$

It proves the same criterion for the entire actual r-torsion set, every natural r. It also checks $|K| = |F|^{[K:F]}$ and derives $[K : F] = k$ from $|K| = |F|^k$, giving the explicit cardinality formulation used by F_(q^k). No rationality predicate or torsion basis is assumed. [pcc:unverified]

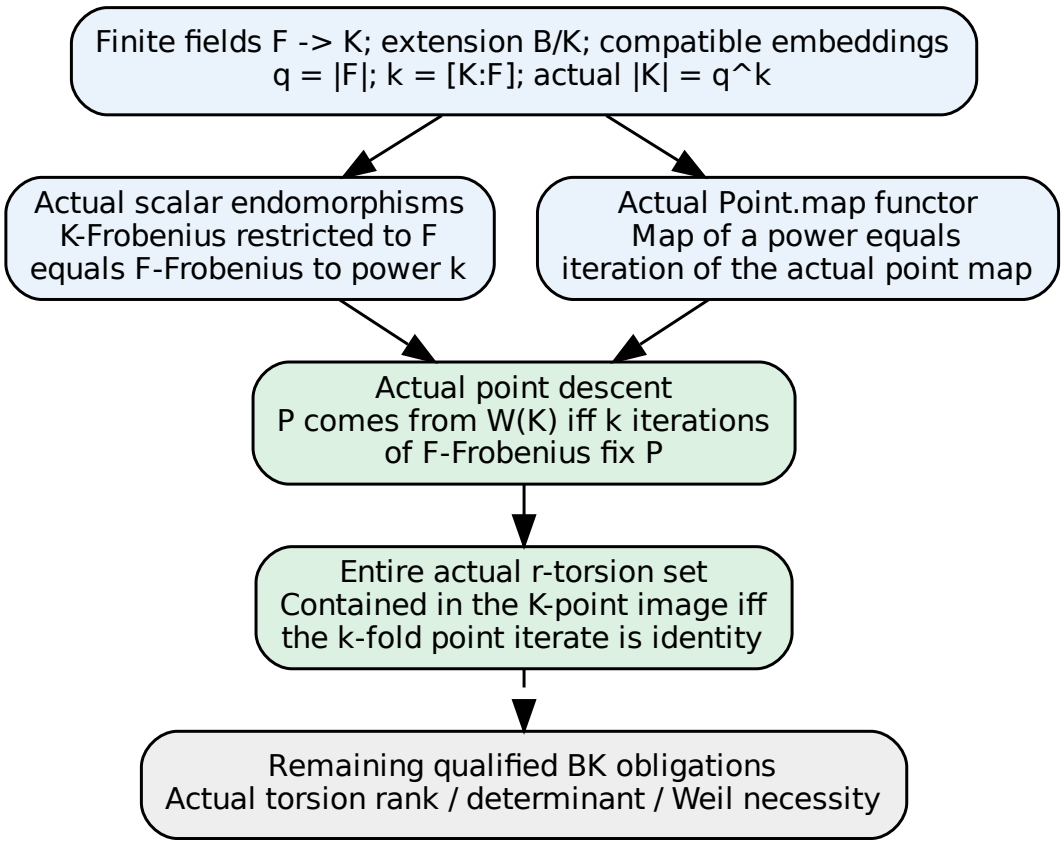


Figure 1: Actual field, point-map and descent dependencies. [pcc:unverified]

What closes the bridge

Actual Point.map respects algebra-homomorphism powers, by its composition law and induction. The actual scalar F-Frobenius to power $[K : F]$ equals K-Frobenius restricted to F, since both act as $x \mapsto x^{|K|}$. Generalized actual finite-base point descent then turns fixedness into an actual W(K) point witness, using coordinate descent and checked nonsingularity transport. Zero is handled separately. Applying this point theorem to every point killed by r gives the torsion-set result. [pcc:unverified]

The chosen embeddings and their compatibility are actual algebra/scalar tower data. This does not construct a canonical finite subfield inside a fixed algebraic closure. Mathlib points contain nonsingular affine points and zero; the descent theorem thus also applies without requiring the whole model's discriminant nonzero. In particular it applies to nonsingular elliptic curves. Torsion is the actual geometric point set, without a group-scheme interpretation. [pcc:unverified]

Main binding: `FrobeniusFiniteExtensionAudit.finite_extension_audit_certificate`. Proof map: `../../FROBENIUS_FINITE_EXTENSION.md`. Exact hypotheses, source hashes, gate and actual packet standing are in `../../frobenius-finite-extension-sources.json` and the separate verification record. Whole census gate490passes, zero failures/zero trusted-native exceptions. Local source commit `3c660a40` precedes packet `PCC-20261110-a496b9`; actual local verdict and signed standing are recorded separately. Lean4.35.0/Mathlib56a00585 are pinned. Only propext, Classical.choice and Quot.sound; no hole, custom axiom, native-decision exception or linter waiver. Failed attempts are preserved; fresh lean4checker is unavailable. This module adds no elaboration option. [pcc:unverified]

Read BK1998 Theorem1/proof/Remark1 pp142–143, DOI10.1007/s001459900040; Mathlib actual Point.map composition, coordinate/nonsingularity descent, AlgHom powers/restriction, finite-field Frobenius, actual cardinality/ finrank and power injectivity. KB/Conductor unavailable; bounded primary/repository/Mathlib fallback, no novelty or broad source promotion. [pcc:unverified]

The earlier nonsingular F7/nonfixed3-torsion disproof remains unchanged: it attacks dropping BK's exclusion, which the qualified theorem and corpus node retain. The new component proves a valid bridge. Actual geometric torsion rank two, determinant q and Weil necessity remain open, as does the all-p<500 census. No native experiment, timing or paid compute. Signed PCC standing remains unverified pending a trusted signed verdict. [pcc:unverified]